

Phase 2 - Observability Setup

Step 2.1 — Redirect Flask Logs to File



Your Setup: Flask app at `/opt/rdsapp/ app.py` running as systemd service `rdsapp`. By default, logs go to journald. CloudWatch Agent reads files more easily, so we redirect output.

Create the log directory

```
sudo mkdir -p /var/log/rdsapp
```

Edit the systemd service

```
sudo nano /etc/systemd/system/rdsapp.service
```

Add these lines under `[Service]` section:

```
StandardOutput=append:/var/log/rdsapp/app.log  
StandardError=append:/var/log/rdsapp/app.log
```

Full `[Service]` section should look like:

```
[Service]  
WorkingDirectory=/opt/rdsapp  
Environment=SECRET_ID=lab/rds/mysql  
ExecStart=/usr/bin/python3 /opt/rdsapp/app.py  
StandardOutput=append:/var/log/rdsapp/app.log  
StandardError=append:/var/log/rdsapp/app.log
```

Reload and restart

```
sudo systemctl daemon-reload
sudo systemctl restart rdsapp
```

Verify logs are flowing

```
sudo tail -f /var/log/rdsapp/app.log
```



Expected: Flask startup messages appear. Press `Ctrl+C` to exit.

Step 2.2 — Add ERROR Logging to `app.py`



CRITICAL: Without explicit `ERROR` in your logs, the CloudWatch alarm won't trigger. This is the most common Phase 4 failure.

Edit the app

```
sudo nano /opt/rdsapp/app.py
```

Replace the `get_conn()` function with:

```
def get_conn():
    try:
        c = get_db_creds()
        host = c["host"]
        user = c["username"]
        password = c["password"]
        port = int(c.get("port", 3306))
        db = c.get("dbname", "labdb")
        return pymysql.connect(host=host, user=user, password=
password, port=port, database=db, autocommit=True)
```

```
except Exception as e:
    print(f"ERROR: Database connection failed - {e}")
    raise
```

Restart the app

```
sudo systemctl restart rdsapp
```

Step 2.3 — Install & Configure CloudWatch Agent

Install the agent

```
sudo yum install -y amazon-cloudwatch-agent
```

Create the config file

```
sudo nano /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwa
tch-agent.json
```

Paste this configuration:

```
{
  "logs": {
    "logs_collected": {
      "files": {
        "collect_list": [
          {
            "file_path": "/var/log/rdsapp/app.log",
            "log_group_name": "/aws/ec2/lab-rds-app",
            "log_stream_name": "{instance_id}",
            "timezone": "UTC"
          }
        ]
      }
    }
  }
}
```

```
}  
}  
}  
}
```

Start the agent

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \  
-a fetch-config \  
-m ec2 \  
-c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json \  
-s
```

Verify agent is running

```
sudo systemctl status amazon-cloudwatch-agent
```

Step 2.4 — Add IAM Permissions for CloudWatch Logs

Option A: Attach managed policy

1. IAM Console → Roles → Your EC2 role
2. Add permissions → Attach policies
3. Search for `CloudWatchAgentServerPolicy`
4. Attach

Option B: Inline policy

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Effect": "Allow",  
      "Action": "logs:CreateLogGroup",  
      "Resource": "arn:aws:logs:*:*:log-group:/aws/elasticbeanstalk/*",  
      "Condition": {"Bool": {"aws:MultiFactorAuthPresent": "false"}},  
      "Sid": "AllowLogsCreateLogGroup" }  
    ]  
}
```

```
{
  "Effect": "Allow",
  "Action": [
    "logs:CreateLogGroup",
    "logs:CreateLogStream",
    "logs:PutLogEvents",
    "logs:DescribeLogStreams",
    "logs:DescribeLogGroups"
  ],
  "Resource": [
    "arn:aws:logs:us-east-2:<ACCOUNT_ID>:log-group:/aws/ec2/lab-rds-app:*",
    "arn:aws:logs:us-east-2:<ACCOUNT_ID>:log-group:*"
  ]
}
```



Why two resources? `DescribeLogGroups` operates at the account level, not the log-group level. The second resource covers this.

Step 2.5 — Verify Log Group Appears

Wait 2-3 minutes, then run:

```
aws logs describe-log-groups \
  --log-group-name-prefix /aws/ec2/lab-rds-app
```



Expected: Log group `/aws/ec2/lab-rds-app` appears in output.

Common Issues — Phase 2

▼ Issue: CloudWatch Agent won't start

Diagnostic:

```
sudo tail -f /opt/aws/amazon-cloudwatch-agent/logs/amazon-cloudwatch-agent.log
```

Common Causes:

- JSON syntax error in config file (missing comma, bracket)
- File path doesn't exist (`/var/log/rdsapp/app.log`)
- Missing IAM permissions

▼ Issue: Log group doesn't appear

Diagnostic:

```
sudo systemctl status amazon-cloudwatch-agent
```

Common Causes:

- Agent not running
- Wrong `file_path` in config
- Missing `logs:CreateLogGroup` permission
- Log file is empty (app not generating logs)

▼ Issue: Agent running but no logs in CloudWatch

Check the local log file exists and has content:

```
ls -la /var/log/rdsapp/  
cat /var/log/rdsapp/app.log
```

If empty, your systemd redirect isn't working. Re-check the service file.

Cleanup Commands (If You Need to Start Over)

```
# Stop and remove agent
sudo systemctl stop amazon-cloudwatch-agent
sudo yum remove -y amazon-cloudwatch-agent
sudo rm -rf /opt/aws/amazon-cloudwatch-agent

# Remove log directory
sudo rm -rf /var/log/rdsapp

# Undo systemd changes
sudo nano /etc/systemd/system/rdsapp.service
# Remove StandardOutput/StandardError lines
sudo systemctl daemon-reload
sudo systemctl restart rdsapp
```

In AWS Console:

- Delete log group `/aws/ec2/lab-rds-app`
- Remove `CloudWatchAgentServerPolicy` from IAM role

Phase 2 Checklist

- ☐ Log directory `/var/log/rdsapp/` created
- ☐ systemd service redirects output to log file
- ☐ `app.py` has `ERROR:` in connection failure logging
- ☐ CloudWatch Agent installed
- ☐ Agent config file created with correct path
- ☐ Agent started and running
- ☐ IAM permissions for CloudWatch Logs attached
- ☐ Log group visible in CloudWatch Console